

ERTS: Assignment 2

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Introduction

The purpose of this assignment is to learn the design flow for a system on a programmable chip. Using Vivado, SDK, and HLS, we explore the relationship between the Zynq Processing System (PS) and Programmable Logic (PL), interface drivers, command-line control, custom IP, and hardware acceleration.

We began by following the setup instructions to configure the ZYBO board, connect it to the development computer, and prepare the required Xilinx design tools. This was initially quite challenging, since our pc runs on Windows 11, and the old software was not fully compatible. Eventually, we got it working. This provided the hardware and software platform for the exercises that follow.

Exercise 3: Create a Console Application

Objective

In this exercise, we create a small console application for the ZYBO board. It receives commands over USB-UART and uses them to control the LEDs through the switches and timer.

Command interpreter

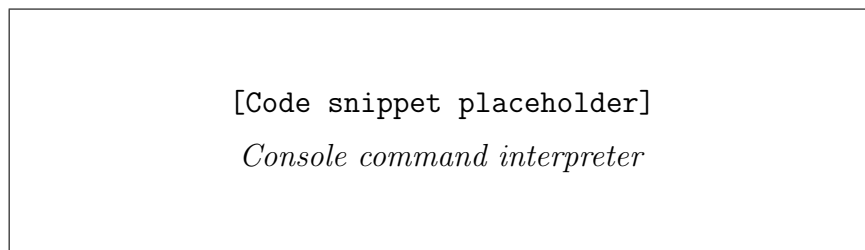


Figure 1: Console command interpreter

The application will read a command from the terminal and use a **switch** statement to select the requested operation.

Command 1: Switches to LEDs

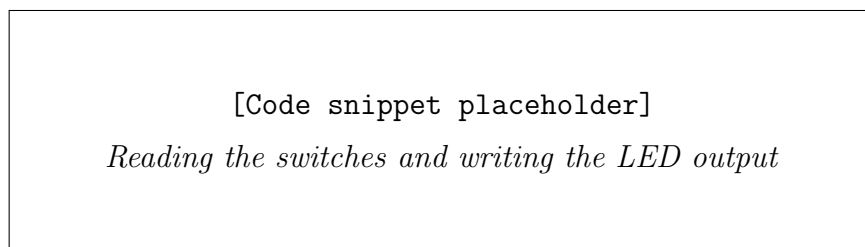


Figure 2: Reading the switches and writing the LED output

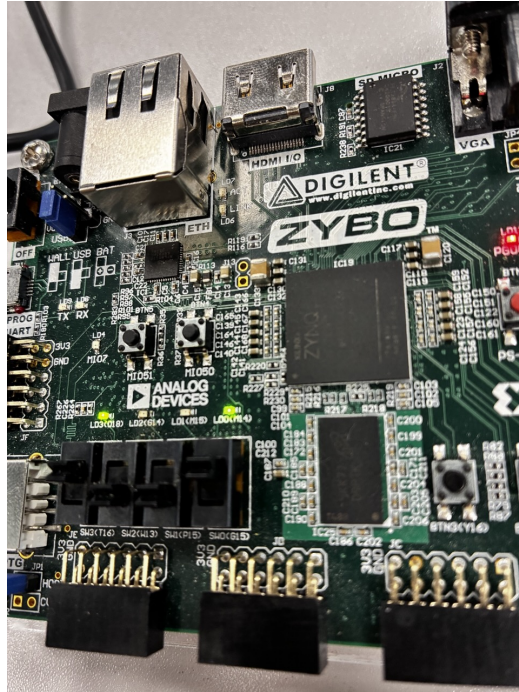


Figure 3: The LEDs reflect the binary state of the switches.

With command 1, the LEDs light up according to the switch positions.

Command 2: Timer-driven binary counter

[Code snippet placeholder]
XScuTimer initialization and one-second counter

Figure 4: XScuTimer initialization and one-second counter

Exercise 4: Create a Matrix Multiplication

Exercise 5: Create a Hardware IP Acceleration

Exercise 6: HLS exercise

Exercise 7: HLS exercise implementing a Neural Network